Name- Chaitanya V. Jhade
Roll no - 48
Batch - T13

THADOMAL SHAHANI TSEC ENGINEERING COLLEGE

SEPM Case-Study

Understanding DevOps - Principles, Practices, and the Role of a DevOps Engineer.

1. Introduction

In today's fast-paced software industry, organizations must deliver high-quality applications rapidly while maintaining reliability and scalability. Traditional software development models often created silos between development and operations teams, leading to delays, inefficiencies and frequent failures in deploy

This case study explores how adopting DevOps practices helped a mid-sized company transform its software delivery lifecycle. It also explains the core principles, practices and responsibilities of a DevOps Engineer in detail

2. Background of the Case

Company Name : TechFusion solutions
Domain : E-commerce platform
Problem statement :
Techfusion faced several issues -

- Slow release cycle
- High failure rate during releases
- Lack of collaboration between development and op
- Manual deployment processes leading to errors
- Difficulty in scaling infrastructure.

3. Challenges Identified
1. Siloed Teams
2. Manual processes
3. Lack of Continuous testing
4. Poor monitoring

4. DevOps adoption strategy
Techfusion decided to adopt DevOps with the following goals:
- Automate workflows
- Improve collaboration
- Enable faster and reliable deployments.
- Enhance system monitoring

5. DevOps Principles
1. Collaboration and Communication
- Breakdown barriers between development and operations teams.
- Encourage shared responsibility.
2. Automation
- Automate repetitive tasks such as testing, building and deployment
3. Continuous Integration (CI)
- Developers frequently merge code into a shared
- Automated builds and tests are triggered
4. Continuous Delivery / Deployment
- Code changes are automatically prepared prepared for release.
- Enables frequent and reliable deployments.

5. Monitoring and Feedback
- Continuous monitoring of application and infrastructure
- Feedback loop helps in quick issue resolution
6. Infrastructure as Code (IaC)
- Infrastructure is managed using code instead of manual configurations.

6. DevOps Practices Implemented
1. CI/CD pipeline
2. Containerization
3. Container orchestration
4. Configuration Management
5. Monitoring and Logging
6. Version control.

7. DevOps Workflow
1. Developer commits code to repository
2. CI Tools automatically builds
3. Doctor Creates Container image
4. Kubernetes deploys Container to production
5. Monitoring tools track performance
6. Feed Back is sent back to developer.

8. Role and Responsibilities of a DevOps Engineer
1. Automation Expert
2. Infrastructure Management
3. Collaboration facilitator
4. Monitoring and Troubleshooting
5. Security Integration
6. Continuous Improvement

9. Result Achieved
- Deployment frequency: Increased from monthly to daily.
- Failure rate: Reduced by 60%
- Deployment time: Reduced by 70%
- System uptime: Improved significantly.

10. Takeaways:
- DevOps is not just a toolset but a cultural shift
- Automation and collaboration are the backbone
- Continuous monitoring ensures reliability
- DevOps engineers play a crucial role in bridging development and operations.